

INSERT TITLE OF YOUR WORKSHOP HERE

INSERT LIST OF ORGANIZERS HERE in the format: name1 (affiliation1),
name2 (affiliation2), . . .

INSERT Start Date–End Date HERE

Start writing here!

Here are some suggestions for topics for your sections:

1 Overview of the Field

1.1 Gaussian process

From a simplified (and likely erroneous?) point of view, a (finite-N-dimensional projection) of a Gaussian process f_N is defined as:

$$f_N \sim \text{multinormal}(\mathbf{m}, \mathbf{K})$$

with a certain mean function $\mathbf{m}$ and a covariance matrix $\mathbf{K}$. In particular, the elements of $\mathbf{K}$ are defined by a covariance function. In our case (and in most Gaussian process applications), we use an *exponentiated quadratic* function:

$$\forall i, j \in \llbracket 1, N \rrbracket \quad k_{i,j} = \gamma^2 \exp \left(-\frac{1}{2} \left(\frac{|x_j - x_i|}{\rho} \right)^2 \right)$$

This covariance function captures the "link" between observations along a continuous variable, such as time or space. The function is characterized by two parameters:

- a marginal deviation γ , which controls the variability around the mean (the intensity?)
- a length scale ρ , which controls how long/far the correlations persist

1.2 Application to tree growth

Now, let's say that we have ringwidth measurements r_w for one tree, for each year x_n . Using one Gaussian process, a simple model would look like this:

$$\forall x_n \quad \log r_w(x_n) \sim \text{normal}(\mu + f(x_n), \sigma^2)$$

with:

$$\vec{f} \sim \text{multinormal}(\vec{0}, \mathbf{K})$$

The two equations are equivalent to:

$$\log \vec{r_w} \sim \text{multinormal}(\vec{\mu}, \Sigma + \mathbf{K})$$

where:

$$\vec{\mu} = \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma^2 \end{bmatrix}$$

In practice, we choose to model the log of ringwidth with two Gaussian processes:

- one **long-term scale** Gaussian process f_{long} , to capture the long-term variation of tree growth, likely related to allometric constraints
- one **short-term scale** Gaussian process $f_{\text{short},p}$ per stand p , to capture some climatic variations over several years

The long-term scale GP is characterized by one pair of parameters $\{\gamma_{\text{sp}}, \rho_{\text{sp}}\}$ per species sp . The short-term scale GP is characterized by two parameters $\{\gamma_{\text{short}}, \rho_{\text{short}}\}$, common across all stands. The short-term scale does not depend on species, but its effect is scaled by a factor κ_{sp} (see below). We chose two different prior models for these two Gaussian processes, to have the prior mass of ρ_{short} between 3 and 10 years, and the prior mass of ρ_{sp} between 20 and 60 years.

The full model is, for n observations for one tree k of the species sp on the stand p :

$$\log \vec{r\tilde{w}}_{sp,p,k} \sim \text{multinormal}(\vec{\mu}_{sp,p}, \Sigma + K_{\text{sp}})$$

with:

$$\log \vec{r\tilde{w}}_{sp,k} = \log \circ \begin{bmatrix} \text{rw}_{sp,p,k}(t_1) \\ \text{rw}_{sp,p,k}(t_2) \\ \vdots \\ \text{rw}_{sp,k}(t_n) \end{bmatrix} \quad \log \vec{\mu}_{sp,p} = \log \circ \begin{bmatrix} \mu_{sp,p}(t_1) \\ \mu_{sp,p}(t_2) \\ \vdots \\ \mu_{sp,p}(t_n) \end{bmatrix}$$

$$\forall i \in \llbracket 1, n \rrbracket \quad \mu_{sp,p}(t_i) = \alpha + \beta_{\text{GSL}} * \text{GSL}(p, t_i) + \beta_{\text{GDD}} * \text{GDD}(p, t_i) + \beta_{\text{SM}} * \text{SM}(p, t_i) + \kappa_{\text{sp}} * f_{\text{short},p}(t_i)$$

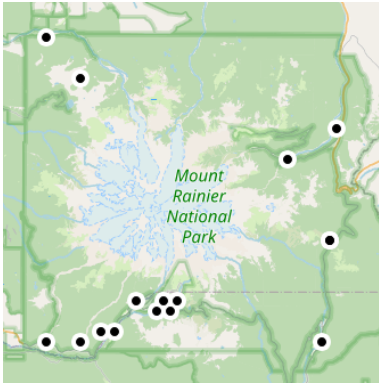
$$\vec{f}_{\text{short},p} \sim \text{multinormal}(\vec{0}, K_{\text{short}})$$

We look at three different climate predictors, at the stand scale:

- the growing season length $\text{GSL}(p, t_i)$, in days
- the growing degree days $\text{GDD}(p, t_i)$ accumulated during the growing season, in $\times 10^\circ \text{C}$
- the mean soil moisture $\text{SM}(p, t_i)$ (weighted average across soil horizons) during the growing season, in %

2 Recent Developments and Open Problems

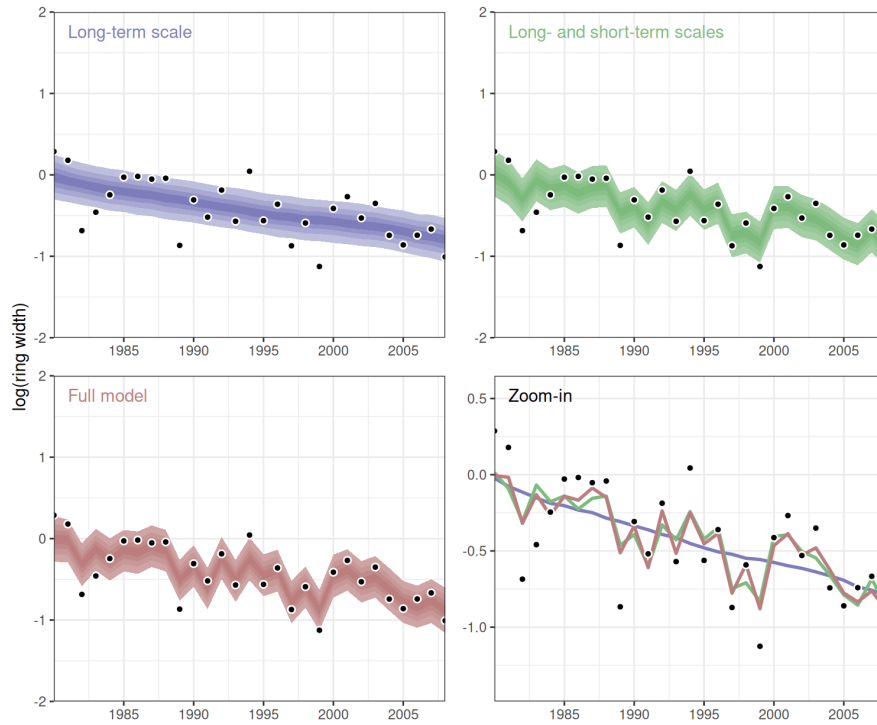
2.1 Study area



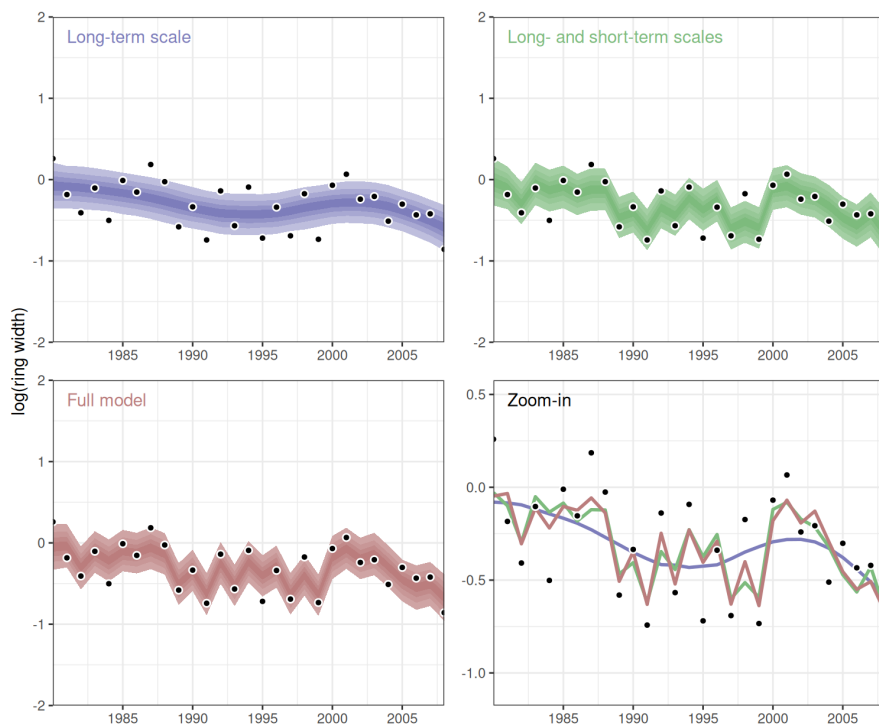
2.2 Retrodictive check

We can examine how the model fit some trees in one particular stand:

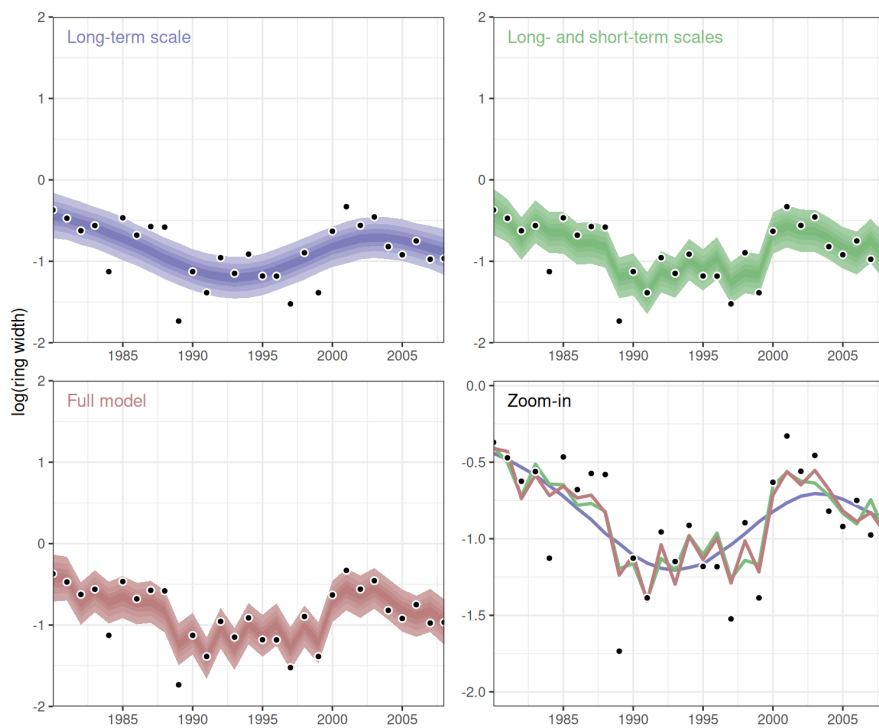
StandAE10, TreeAE10_F1



StandAE10, TreeAE10_F10



StandAE10, TreeAE10_F12



- 3 Presentation Highlights**
- 4 Scientific Progress Made**
- 5 Outcome of the Meeting**